

THERMODYNAMICS 1(MEC 122) LECTURE NOTES

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FUNDAMENTAL CONCEPTS AND LAWS OF THERMODYNAMICS

Introduction

Thermodynamics is a branch of mechanical engineering concerned with the relationship between heat, work, energy, and temperature of engineering systems and those properties of matter that relate to heat and work. In a broader form, it is a science in which energy storage, transformation, and transfer are studied. Energy is stored as internal energy (associated with temperature), kinetic energy (due to motion), potential energy (due to elevation), and chemical energy (due to chemical composition). It is transformed from one of these forms to another and transferred across a boundary as either heat or work.

Mathematical equations are developed in thermodynamics that relate energy transformations and transfers to material properties such as temperature, pressure, or enthalpy. Substances and their properties thus become an essential secondary subject. Much of our work will be based on experimental observations organized into mathematical statements or laws; thermodynamics' first and second laws are the most widely used.

When engineers study thermodynamics, their main goal is frequently to analyze or build a large-scale system, such as a nuclear power plant or an air conditioner. The activity of the constituent molecules in such a system can be averaged into quantifiable variables like pressure, temperature, and velocity [1].

Thermodynamics comprises two words: **thermo** (heat) and **dynamics** (power/transfer of heat energy). Therefore, **Thermodynamics** is concerned with the interactions of a system and its surroundings or one system interacting with another. A system interacts with its surroundings by transferring energy across its boundary. Whenever there is an interaction between energy and matter, thermodynamics is involved.

Area of application: In terms of scope of application, thermodynamics is commonly encountered everywhere in life, such as in the pumping of blood in your

body. **Household applications** include cooking stoves, pressure cookers, Air conditioning, iron, TV, Mobile, heater, refrigerators, etc. **Industrial applications** are automotive vehicles, airplanes, jet engines, nuclear power plants, Alternative Energy (solar, wind, Biomass), cooling, heating systems etc. Thermodynamics entails four laws known as Zeroth, First, Second, and Third laws of thermodynamics.

Thermodynamics entails four laws or axioms: Zeroth, First, Second and Third laws of thermodynamics.

1. The Zeroth law deals with thermal equilibrium and establishes a concept of temperature.
2. The First law throws light on the concept of internal energy.
3. The Second law indicates the limit of converting heat into work and introduces the principle of increase of entropy.
4. The Third law defines the absolute zero of entropy.

These laws are based on experimental observations and have no mathematical proof. Like all physical laws, these laws are based on logical reasoning [2].

SYSTEM, BOUNDARY, SURROUNDINGS WORKING FLUID OF A SYSTEM

A thermodynamic **system** is a quantity of matter or a region within a closed boundary (surface). The surface that separates the system from its surroundings is called the **boundary**. The mass or area outside the system is called the **surroundings** (Figure.1). **The Boundary** of a system can be a **real** one like that enclosing the gas in the cylinder of Figure.1 or an **imaginary** boundary like that of a nozzle opened from both ends (Figure 6). In the case of a nozzle, the inner surface of the nozzle forms the **real** part of the boundary, and the entrance and exit areas form the **imaginary** part since there are no physical surfaces (Figure 6). The boundary may also be fixed (e.g., potato) or movable (e.g., piston-cylinder). Note that the boundary is the contact surface shared by the system and the surroundings. In Figure.1, **the system is the compressed gas, the working fluid, and the dotted line shows the system boundary.**

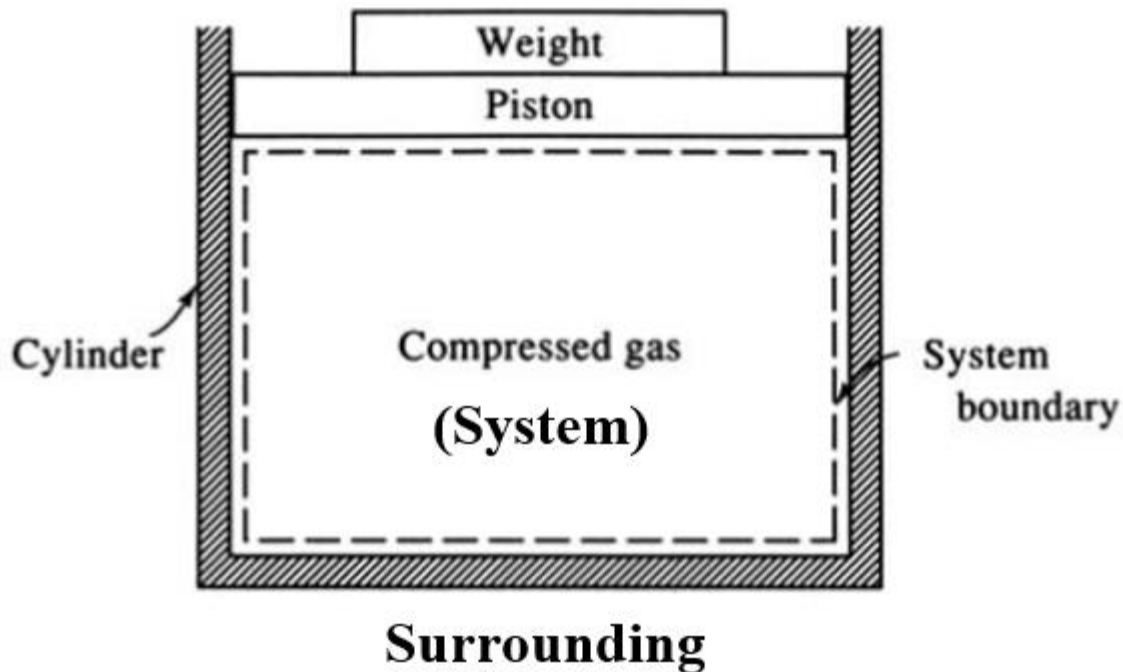


Figure.1 System, surroundings, boundary

A universe comprises a system, its surroundings, and the system boundary. We allow two essential interactions between the system and its surroundings:

- Heat can cross into the system (our potato can get hot), and
- Work can travel out of the system (our potato can expand).

The system boundaries can change; for example, the potato might expand on heating, but we can still distinguish the system and the surroundings.

The working fluid in a thermodynamic system is a liquid or gas that absorbs or transmits energy. The working fluid can be described as the **matter contained within the boundaries of a system**. In other words, it is a transport agent for energy and mass interactions. The working fluid can either be vapor or gaseous [3]. The working fluid used in the refrigeration cycle is called refrigerant. A working fluid can be a pressurized gas or liquid that actuates a machine. Examples include steam in a steam engine, air in a hot air engine, and hydraulic fluid in a hydraulic motor or cylinder. Heat engines and other cyclic devices usually involve a fluid to and from

which heat is transferred while undergoing a cycle. The state of the working fluid is defined by specific characteristics known as properties [1].

TYPES OF THE THERMODYNAMIC SYSTEMS

After defining the system and boundary, observing the interaction between the system and its surrounding across the boundary is crucial. This interaction can be in the form of mass and energy. Based on the interaction across the boundary, the system can be categorized into three types: **closed**, **isolated**, and **opened** (Figure.2), depending on whether a fixed mass or a fixed volume in space is chosen for study.

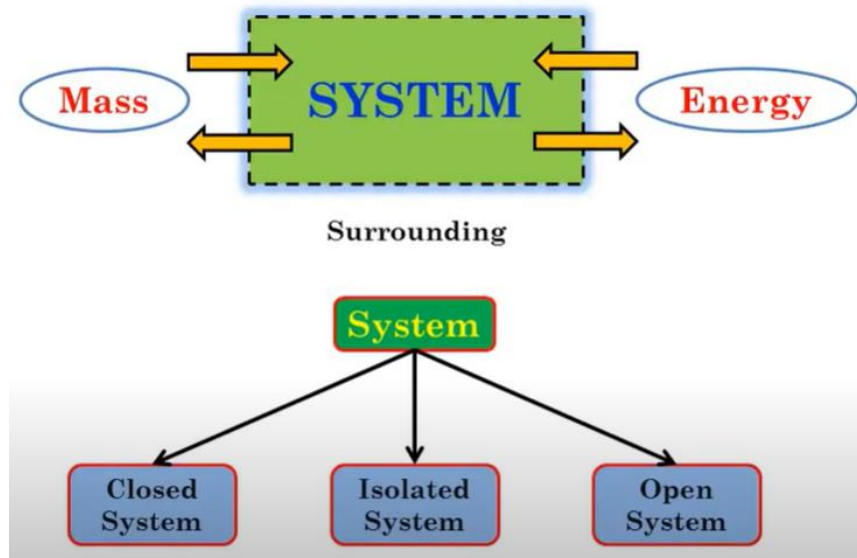


Figure.2 Types of the thermodynamic system in detail

i. Closed system (control mass)

A closed system is a type of system that has only energy interaction with the surroundings but no mass interaction. For example, consider a tightly closed cylinder device (Figure 3). In this system, no mass can transfer in or out of the boundary, but energy can move across the system boundary in the form of Heat or Work. It is also known as **control mass** because it consists of a fixed amount of mass, and no mass can cross its boundary. Note that the volume of a closed system does not have to be fixed.

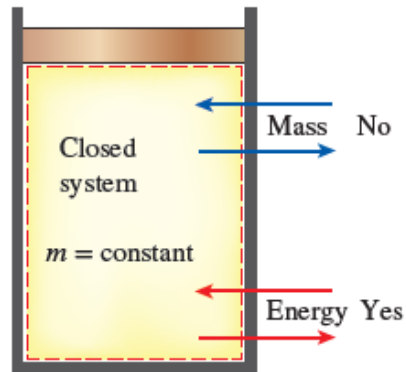


Figure 3 A closed system mass cannot cross the boundaries of a closed system, but energy can

An example of a closed system with a moving boundary considers a tight piston-cylinder device (without valves), as shown in Figure 4. Let's say we would like to find out what happens to the enclosed gas when it is heated. Gas inside the cylinder is our system. The inner surface of the piston cylinder forms the boundary. It is interesting to observe that this boundary is a moving boundary and the real boundary, and since no mass crosses the boundary, this is called a closed system. Note that here energy is crossing the boundary in the form of heat, and the part of the boundary may move because of the expansion of the gas. Everything outside the gas, including the piston cylinder, is called the surrounding.

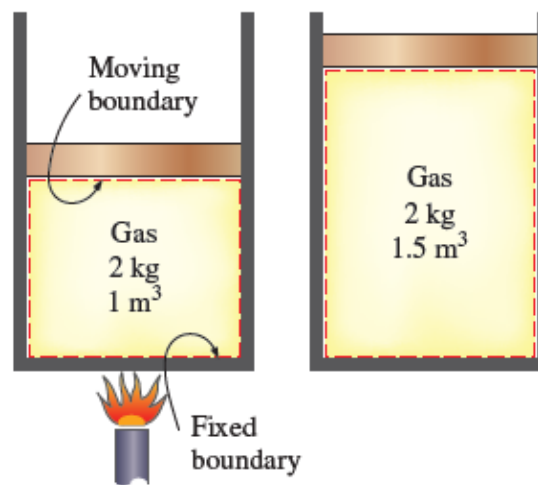


Figure.4 A closed system with a moving boundary

Keep some amount of fluid in the cylinder and close all the valves. Applying heat to the bottom of the cylinder will start to expand the fluid (Figure 4), and as a result, the temperature of the fluid will gradually increase, and that heat will be emitted outside. So, we added energy into the system and energy was transferred too. Hence, there is energy transfer, although there is no mass transfer. Other examples include water in a closed pressure vessel, cooking in a pressure cooker, a Car Battery, a Piston cylinder, and a can of cold drinks.

ii. Isolated system

Isolation means being separated from something. An isolated system is not influenced by its surroundings, so there will be no mass or energy transfer between it and its surroundings (Figure.5). Thermos Flask is an example of an isolated system. We all have flasks in our homes.

- a. What do we do with it? We stored hot water, hot tea, or hot milk for a long time.
- b. How is it possible? Because this flask is well **insulated**, it helps prevent heat energy from going out, and we get hot water.
- c. But have you observed that the temperature of hot water after a long time is reduced and will be the same as normal temperature? Why? Because the insulation is not perfect; if the insulation is perfect, it will not allow any heat transfer across the boundary; then, there will not be any mass transfer or energy transfer. Thus, **an isolated system is a special case of a closed system.**

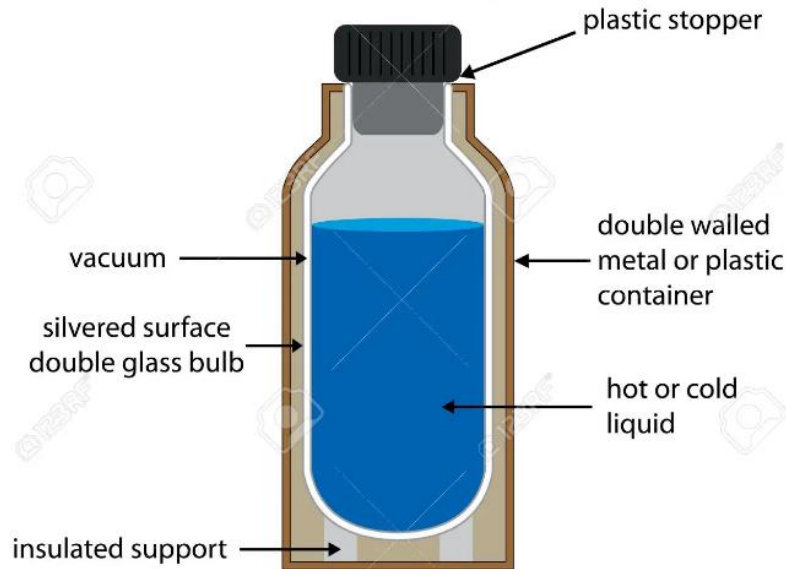
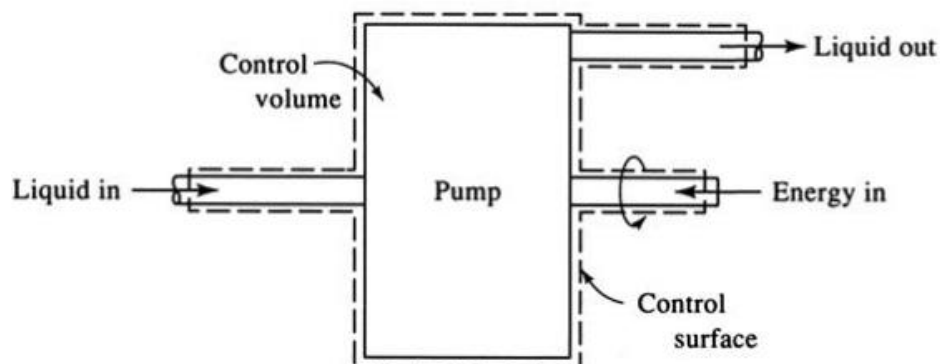


Figure.5 An Isolated system of thermos Flasks

iii. Open system (control volume)

Refer to Figure 6. An open system is one in which matter flows into or out of the system. Most of the engineering systems are open, such as a pump, compressor, Boiler, turbine, or nozzle, water in a pipe, a water heater, Engine oil in the engine, car radiator, pump, and liquid in the heat exchanger. An open system, also known as control volume.

, A control volume can also involve heat and work interactions just as a closed system, in addition to mass interaction.



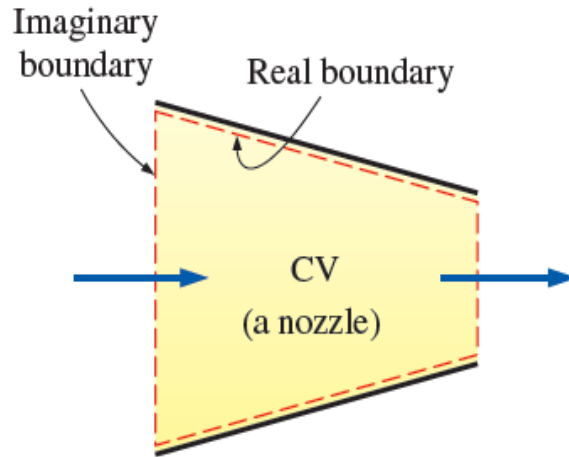


Figure 6 A control volume (CV) with real and imaginary boundaries

THERMODYNAMIC STATE, EQUILIBRIUM,

State' of a system indicates the **specific condition** of the system as described by its properties. At a given state, all the properties of a system have fixed values. Thus, if the value of even one property changes, the state will change to a different one.

A thermodynamic system changes due to energy and mass interactions. The thermodynamic state of the system changes due to these interactions. The mode in which a system's change of state occurs is termed as the **process** such as constant pressure process, constant volume process etc. The succession of states passed through during the change of state is called the **path of the change of state**. Thus, a process transforms from one state to another (Figure 7), e.g., a constant pressure process. When all the properties of a system have definite values, the system is said to exist at a **definite state**.

In an **equilibrium** state, the system has no unbalanced potentials (or driving forces). A system in equilibrium experiences no changes when it is isolated from its surroundings.

Types of equilibrium

- i. **Thermal equilibrium:** when the temperature is the same throughout the entire system.

- ii. **Mechanical equilibrium:** when there is no change in pressure at any point of the system. However, the pressure may vary within the system due to gravitational effects.
- iii. **Phase equilibrium:** in a two-phase system, when the mass of each phase reaches an equilibrium level.
- iv. **Chemical equilibrium:** when the chemical composition of a system does not change with time, i.e., no chemical reactions occur.

PROCESS AND CYCLE

Any change a system undergoes from one equilibrium state to another is called a **process**, and the series of states through which a system passes during a process is called a path.

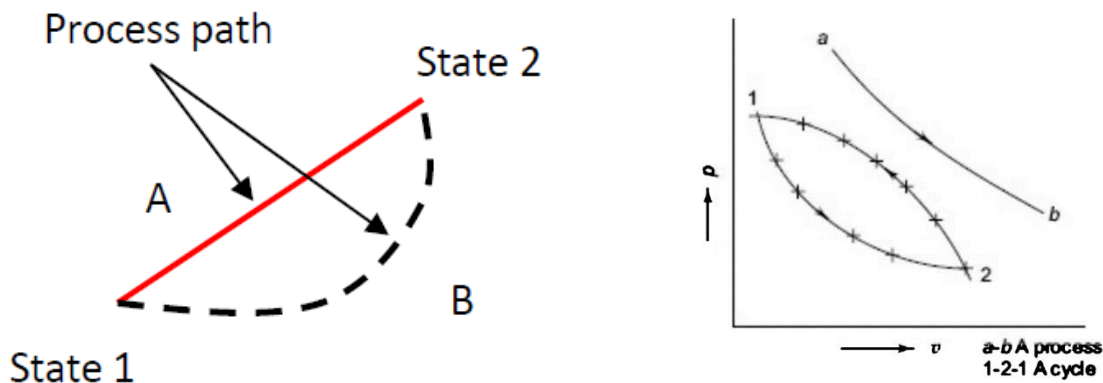


Figure 7 A state, process and a cycle

Note: Initial and final states and path must be specified to specify a process.

Any process or series of processes with identical end states is termed a **cycle**. Thus, a **thermodynamic cycle** is a series of processes (state changes) in which a working fluid begins at an initial state, moves through one or more intermediate states, and then returns to its initial state (Figure 7). If the network is produced by the cycle, it is labeled a power cycle. If work is added to the cycle and heat is pumped from a cold space to a hot space, the cycle is refrigerated.

A closed system is said to undergo a cyclic process, or cycle when it passes through a series of states so that its final state is equal to its initial state.

QUASI-EQUILIBRIUM PROCESS

Quasi-static (Quasi-equilibrium): Quasi means 'almost'. A quasi-static process is also called a reversible process. It can be viewed as a sufficiently slow process that allows the system to adjust itself internally and remains infinitesimally close to an equilibrium state at all times. Quasi-equilibrium process is an idealized process and is not a true representation of the actual process. We model actual processes with quasi-equilibrium ones. Moreover, they serve as standards for comparing actual processes [2].

To consider how a gas (or liquid) might be expanded or compressed in a quasi-equilibrium fashion, refer to Figure.8, which shows a system consisting of a gas initially at an equilibrium state. As shown in the figure, the gas pressure is maintained uniformly throughout by several small masses resting on the freely moving piston. Imagine that one of the masses is removed, allowing the piston to move upward as the gas expands slightly. During such an expansion, the state of the gas would depart only slightly from equilibrium. The system would eventually come to a new equilibrium state, where the pressure and all other intensive properties would be uniform in value again.

Moreover, if the mass were replaced, the gas would be restored to its initial state, while the departure from equilibrium would be slight. If several of the masses were removed one after another, the gas would pass through a sequence of equilibrium states without ever being far from equilibrium. As the mass increments vanish in the limit, the gas undergoes a quasi-equilibrium expansion process. A quasi-equilibrium compression can be visualized with similar considerations.

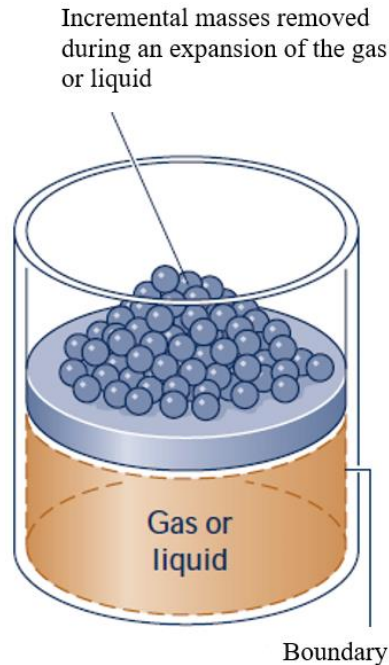


Figure.8 Illustration of a quasi-equilibrium expansion or compression

THERMODYNAMICS PROPERTIES

Any system characteristic that can be observed or measured is called a property of the system, e.g., volume, temperature, pressure, etc. Thermodynamic properties are defined as characteristic features of a system capable of specifying the system's state. Some familiar properties are pressure P , temperature T , volume V , and mass m . The list can be extended to include less familiar ones such as viscosity, thermal conductivity, modulus of elasticity, thermal expansion coefficient, electric resistivity, and even velocity and elevation. The thermodynamic properties of the system are further classified as the '**intensive** property' and '**extensive property**.'

Intensive properties

A quantity is called **intensive** when it is independent of the mass of a system or properties which have the same value for any part of the system, such as temperature, pressure, velocity, density, Specific Volume, Specific Entropy, Thermal conductivity, Thermal Expansion, Compressibility and many more. (Figure 9). **They are not additive**

Extensive properties

Extensive properties are those properties that depend upon the mass of the system and do not maintain the same value for any path of the system, such as mass, volume, momentum, Energy (U), enthalpy (H), heat capacity, entropy (S), etc. **They are additive** [3].

An easy way to determine whether a property is intensive or extensive is to divide the system into two equal parts with an imaginary partition, as shown in Figure 9. Each part will have the same value of intensive properties as the original system but half the value of the extensive properties.

- Generally, uppercase letters denote extensive properties (except mass m), and lowercase letters are used for intensive properties (except pressure P and temperature T).
- Extensive properties per unit mass and moles are called **specific properties** e.g., specific volume ($v = V/m$) and specific energy ($e = E/m$). In this case, we speak of a **specific quantity** or a **reduced extensive quantity**.

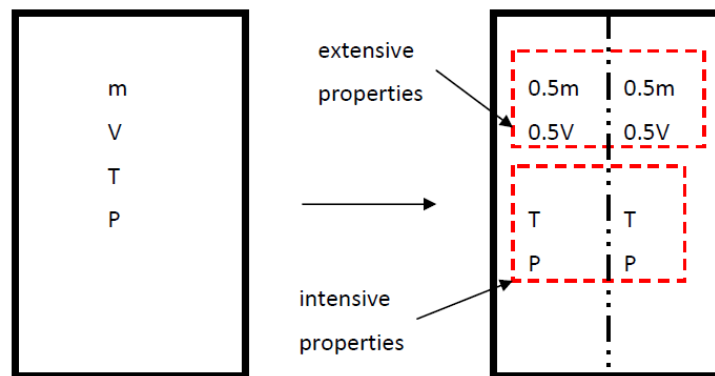


Figure 9 Intensive and extensive properties of a system.

DEFINITION OF THE THERMODYNAMICS PROPERTIES

1. Temperature

The temperature (T) is a thermal state of a body that distinguishes a hot body from a cold body. Thus, temperature is a pointer for the direction of energy transfer as heat.

Instruments for measuring ordinary temperatures are known as thermometers, and those for measuring high temperatures are known as pyrometers.

It has been found that gas will not occupy any volume at a certain temperature. This temperature is known as absolute zero temperature. The temperatures measured with absolute zero as a basis are called **absolute temperatures**.

Absolute temperature is stated in the Kelvin scale. The point of absolute temperature is found to occur at 273.15 K below the freezing point of water. Then, Absolute temperature = Thermometer reading in °C + 273.15. Another scale is the Celsius scale.

The introduction of the Zeroth law of thermodynamics further refined the concept of temperature. It is called this because it was introduced about 100 years after the first and second laws.

2. Pressure

Pressure, P , is the normal force exerted per unit surface area within the system.

$$P = \frac{F}{A} = \frac{F}{A} \text{ Pa} \quad (1)$$

We speak of pressure in fluids, gases and liquids; in solids, this is stress. For a fluid at rest, the pressure at a given point is the same in all directions. See Figure 10.

For engineering work, pressures are often measured concerning atmospheric pressure rather than absolute vacuum.

$$P_{\text{abs}} = P_{\text{atm}} = P_{\text{gauge}} \quad (2)$$

In SI units, the derived unit for pressure is the Pascal (Pa), where $1 \text{ Pa} = \frac{\text{N}}{\text{m}^2}$. This is very small for engineering purposes, so pressures are usually given in terms of kilopascals ($1 \text{ kPa} = 10^3 \text{ Pa}$), megapascals ($1 \text{ MPa} = 10^6 \text{ Pa}$), or bars ($1 \text{ bar} = 10^5 \text{ Pa}$) and the standard atmosphere, where $1 \text{ atm} = 101\,325 \text{ Pa}$.

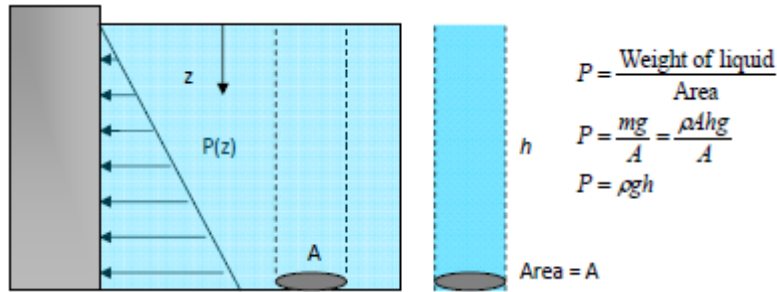


Figure 10 Fluid pressure at rest increases with depth (due to added weight) but is constant in horizontal planes.

3. Specific volume (v) and Density (ρ)

The **specific volume** of a system is the volume occupied by the unit mass of the system. The symbol used is v , and the unit is m^3/kg . The symbol V will be used for volume. (Note that specific volume is reciprocal of density). Mathematically as in equation 3b.

$$v = \frac{\text{volume}}{\text{mass}} \quad (3)$$

Units are m^3/kg

It represents the inverse of the density, $v = \frac{1}{\rho}$ (3b)

4. Internal Energy (U)

Internal energy (U) - The property of a system covering all forms of energy arising from the internal structure of the substance, SI Unit **kJ**. **Specific energy** (u) is $\frac{E}{m}$. SI Unit **kJ/kg**

5. Enthalpy

Enthalpy (H) - A system property conveniently defined as $H = U + PV$, where U is the internal energy. SI Unit **kJ**. **Specific enthalpy**, $h = \frac{H}{m} = u + Pv$, SI Unit **kJ/kg**

6. Entropy

Entropy (S) - The microscopic disorder of the system. It is an extensive equilibrium property. SI Unit **kJ**, but specific entropy (s) become intensive. SI Unit **kJ/kg**. [4]

7. Specific Heats

The **specific heat** is defined as the energy required to raise the temperature of a unit mass of a substance by one degree. In general, this energy depends on how the process is executed. We are interested in two kinds of specific heat in thermodynamics: **specific heat at constant volume**. C_v and **specific heat at constant pressure** C_p .

Physically, the specific heat at a constant volume C_v It can be viewed as the energy required to raise the temperature of the unit mass of a substance by one degree as the volume is maintained constant. The energy required to do the same as the pressure is maintained constant is the specific heat at constant pressure. C_p . Specific heat at constant pressure C_p is always greater than C_v Because the system is allowed to expand at constant pressure, the energy for this expansion work must also be supplied to the system.

LAWS OF THERMODYNAMICS

i. Zeroth law of thermodynamics

If an object with a higher temperature comes in contact with a lower-temperature object, it will transfer heat to the lower-temperature object. The objects will approach the same temperature, and without losing to other objects, they will maintain a constant temperature. Therefore, thermal equilibrium is attained.

The "**Zeroth law**" states that if two systems are simultaneously in thermal equilibrium with a third system, they are in thermal equilibrium.

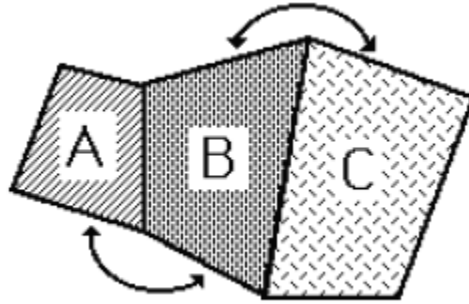


Figure11 Analogy of the Zeroth Law of Thermodynamics

If A and C are in thermal equilibrium with B, then A is in thermal equilibrium with B. Practically, all three are at the same temperature. It forms the basis for comparing temperatures.

In other words, the Zeroth Law states that:-"two systems which are equal in temperature to a third system are equal in temperature to each other." This law deals with thermal equilibrium and establishes a concept of temperature.

ii. The first law of thermodynamics

The first law of thermodynamics is the application of the conservation of energy principle. States that energy can be neither created nor destroyed during a process; it can only change forms. This means that energy can change from one form to another during an interaction, but the total amount of energy remains constant. This law throws light on the concept of internal energy.

Example of first law applicable to a closed system. Consider a closed system with no flow into or out of the system, and the fluid mass remains constant. For such a system, the first law statement is known as the Non-Flow Energy Equation, or NFEE abbreviated, it can be summarised as follows:

$$\Delta U = Q - W \quad 4$$

The first law makes use of the key concepts of internal energy (ΔU), heat (Q), and system work (W).

iii. The second law of thermodynamics

The second law of thermodynamics states that heat will flow naturally from one energy reservoir to another at a lower temperature but not in the opposite direction without assistance. This is important because a heat engine operates between two energy reservoirs at different temperatures. Further, the first law of thermodynamics

establishes equivalence between the quantity of heat used and the mechanical work. Still, it does not specify the conditions under which conversion of heat into work is possible, nor the direction in which heat transfer can occur. The second law of thermodynamics has bridged this gap.

v. Third Law of Thermodynamics

The third law of thermodynamics states, "When a system is at zero absolute temperature, the entropy of the system is zero."

All the laws of thermodynamics are based on experimental observations and have no mathematical proof. Like all physical laws, these laws are based on logical reasoning.

QUESTIONS

Briefly answer the following questions:

1. What is thermodynamics? List 6 the application areas of thermodynamics.
2. Write short notes on the following: Thermodynamic properties with 5 examples with their SI units, closed system, isolated system, and open system, extensive and intensive properties. Support your explanation with examples and sketches where necessary.
3. Briefly explain the term working fluid in thermodynamics with examples
4. Explain the following terms: (i) State, (ii) path, (iii) Process, (iv) Cycle and (v) specific heat at constant volume and pressure.
5. State the four laws of thermodynamics.
6. What is a quasi-static process?
7. State thermodynamic definition of work. Also, differentiate between heat and work.
8. What is energy? What are the different forms of it?